

Full length article

IndVisSGG: VLM-based scene graph generation for industrial spatial intelligence

Zuoxu Wang^a, Zhijie Yan^a, Shufei Li^{b,c}*, Jihong Liu^a^a School of Mechanical Engineering and Automation, Beihang University, Beijing, China^b Centre for Advanced Robotics Technology Innovation, Nanyang Technological University, Singapore^c State Key Laboratory of Digital Manufacturing Equipment and Technology, Huazhong University of Science and Technology, Wuhan, China

ARTICLE INFO

Keywords:

Scene graph

Vision-language model

Industrial spatial intelligence

Smart manufacturing

ABSTRACT

Industrial spatial intelligence enables robots and machine tools to understand environmental settings and their relationships, allowing them to manipulate target components. A crucial aspect of this process is scene graph generation (SGG). Previous research on SGG primarily focuses on detection and panoptic segmentation of objects, followed by the prediction of their pairwise relationships. However, these approaches struggle with generalization and transferability when encountering new scenarios. To tackle this problem, we propose the *Industrial Visual Scene Graph Generation* (IndVisSGG) method, which parses spatial and interactive relationships between objects in temporal industrial settings. This approach leverages the capabilities of *Vision-Language Models* (VLMs) to generate scene graphs quickly and accurately without any additional object annotations. Furthermore, leveraging the IndVisSGG method, we have implemented a meticulous annotation procedure to compile a high-quality *industrial scene graph generation* (ISG) dataset, comprising 10,000 images of manufacturing and related industrial scenes. Through comparisons with various scene graph generation methods and benchmarks across two other datasets, we have showcased the superiority of the IndVisSGG method and underscored the benefits of the ISG dataset over existing datasets.

1. Introduction

With the rapid advancement of sensors, intelligent devices, and artificial intelligence algorithms, modern intelligent manufacturing systems have shown considerable success in perception tasks such as object recognition and pattern recognition [1,2]. However, perception intelligence alone is insufficient for creating a flexible and adaptive production environment that aligns with the principles of human-centricity, sustainability, and resilience, as envisioned in Industry 5.0 [3–5]. To realize these goals, industrial spatial intelligence is essential. It empowers robots and machines to comprehend the physical world, make informed production decisions, execute appropriate actions, and interact seamlessly with humans, thereby facilitating flexible automation [6].

To understand the manufacturing scenes, scene graph generation (SGG) is a fundamental technique in industrial spatial intelligence, enabling the parsing of relative positions and associative relationships among diverse industrial parts within a manufacturing workshop [7, 8]. SGG can produce a graph-structured representation of pairwise

object relationships for a given image of on-site industrial settings. This representation can be applied to downstream applications such as visual reasoning in non-structured manufacturing scenarios [9], visual question answering during production processes [10], and task allocation for multiple human–robot collaborations [11].

Nevertheless, existing research on SGG typically relies on multiple stages: visual detection and segmentation to identify objects through bounding boxes and masks, link prediction of object relations for graph construction, and edge classification between node pairs in the graph. An SGG model requires a large amount of annotated visual data to encode image-to-text relations during the training process. Furthermore, pre-trained SGG models often lack generalization and transferability when applied to new industrial tasks, impeding the transition from prototyping to production in scenarios with product variants.

To address these challenges, this paper proposes IndVisSGG, a novel Vision-Language Model-based Scene Graph Generation method that leverages the advanced capabilities of GPT-4V (vision) for quick and

* Corresponding author.

E-mail address: shufei.li@outlook.com (S. Li).<https://doi.org/10.1016/j.aei.2024.103107>

Received 13 August 2024; Received in revised form 6 November 2024; Accepted 31 December 2024

Available online 11 January 2025

1474-0346/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

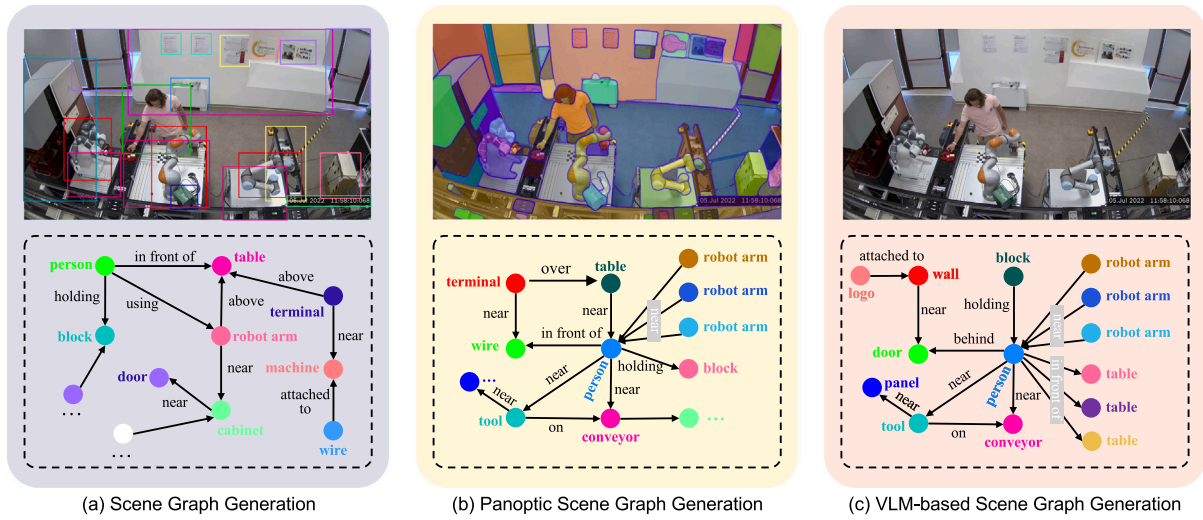


Fig. 1. The comparison between IndVisSGG (c) and other scene graph generation methods (a, b).

adaptive scene interpretation in temporal scenarios. Instead of merely focusing on locating objects, IndVisSGG demonstrates remarkable reasoning capability to generate triples that reflect both positional information and associative relations between various objects in industrial scenarios without requiring additional object annotation. The contributions of this research are as follows:

- IndVisSGG, a portable scene graph generation method based on GPT-4V, is proposed. A multiple artificial intelligence (AI) agent strategy is designed to enhance temporal scene understanding capabilities, thereby improving scene graph generation performance. IndVisSGG paves the way for the use of VLMs in SGG tasks.
- A new SGG dataset, named ISG, has been developed with abundant scene graph annotations. It contains 10,000 well-annotated scenes from the industrial and related manufacturing fields. Compared to existing datasets, ISG is distinguished by its more detailed and precise graph annotations.
- IndVisSGG is evaluated on two representative datasets. The ISG dataset is also used to test two VLMs within the multiple AI agent strategy for scene parsing. The results demonstrate the effectiveness of IndVisSGG and highlight the advantages of the ISG dataset.

The rest of this article is organized as follows. Section 2 introduces relevant studies regarding industrial spatial intelligence, SGG, and large language models (LLMs) in manufacturing settings. Section 3 presents the technical details of the proposed approach, which involves the spatial-temporal triplets extracting, triple-checking, and triplets summarizing. Section 4 elaborates on the development of the industrial scene graph dataset. Section 5 presents the experimental results and demonstrates the proposed method applied in industrial scenarios. Finally, Section 6 concludes this work and highlights future perspectives.

2. Related works

2.1. Industrial spatial intelligence

Industrial spatial intelligence refers to the ability of machines, robots, and systems within industrial environments to understand, interpret, and interact with the spatial aspects of their surroundings [12]. In production environments, which involves humans, machines, and robotic agents [13], industrial spatial intelligence holds even greater significance. It is a prerequisite for tasks such as dexterous robot

manipulation [14], smart assembly [15], and flexible sequential operations [16].

For visual perception in industrial spatial intelligence, Fan et al. [17] proposed a vision-based 3D dense hand-object pose estimation approach to perceive the pose of the human hand and workpiece under partial occlusion. For predictive capability, Li et al. [18] utilized a multimodal transfer learning algorithm to forecast incoming events in human-robot collaborative assembly tasks. For proactive action intelligence, Zhao et al. [19] proposed a hybrid learning method to enable a robot to imitate human dexterous actions. Despite the advantages of these methods, the semantic knowledge of diverse industrial scenarios requires further exploration to achieve comprehensive industrial spatial intelligence.

2.2. Scene graph generation

The SGG technique can learn spatial information and semantic knowledge from visual data, consisting of three sub-steps: object detection, relation detection, and graph construction [20]. Besides, most research achieves SGG through either bounding box detection or panoptic segmentation manner.

Detection-based SGG employs object detection algorithms, such as Faster R-CNN, to identify various objects in images, providing each detected object with a bounding box and a category label [21]. However, several issues should be considered when developing such algorithms. Firstly, bounding boxes marked as annotations may only offer rough localization of objects [22], including noise or blurred pixels belonging to different objects or categories, as seen in the overlapping areas between the blue and yellow bounding boxes in Fig. 1a. Secondly, bounding boxes usually do not cover the entire scene of an image. For example, in Fig. 1a, the road area, crucial for understanding the scene, is completely ignored. Lastly, data annotated based on bounding boxes often contain inconsistent and redundant labels, such as the multiple markings of trees and the sky in Fig. 1a, where some additional annotations contribute nothing to scene understanding [23] (see isolated nodes). Learning from such labels could confuse the model.

Unlike traditional detection-based SGG methods, Panoptic Segmentation-based SGG (PSG) uses mask segmentation instead of bounding boxes to represent objects [24], enabling a more comprehensive understanding, as illustrated in Fig. 1b. It leverages panoptic segmentation algorithms, such as Mask2Former [25], to classify each pixel for precise object labeling. However, SGG results are still limited by the quality of annotations in this manner, and the annotation process is resource-intensive. Inappropriate labeling of object types

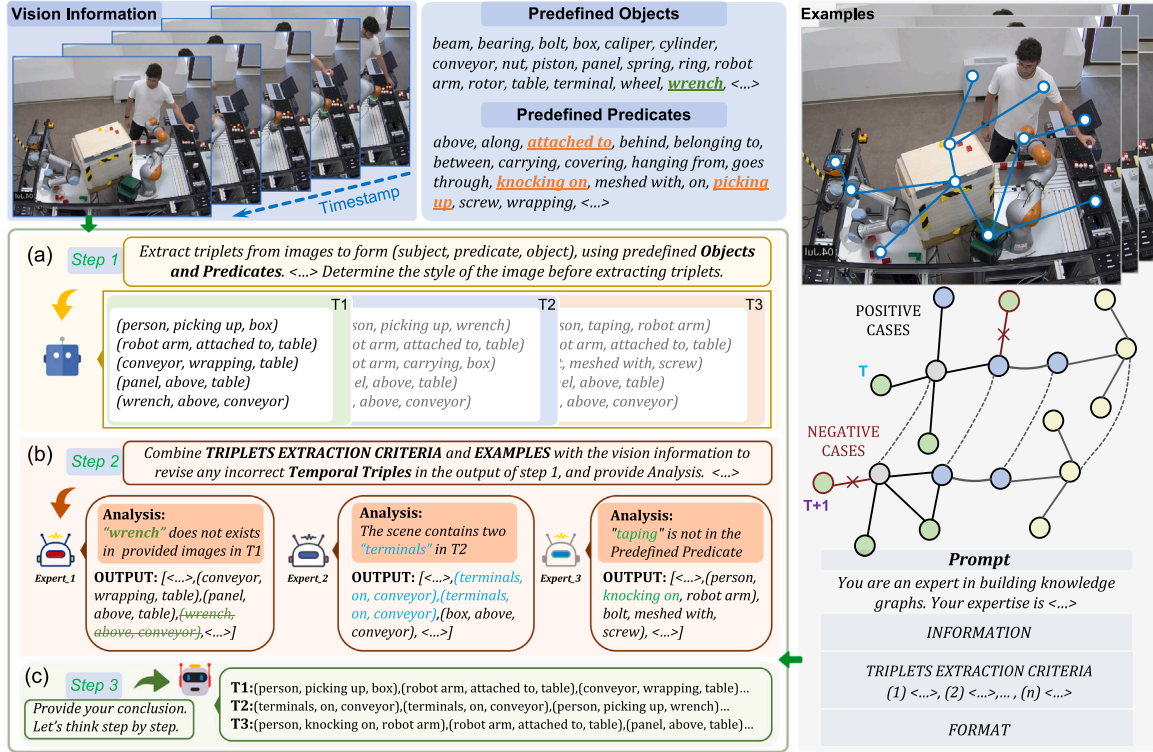


Fig. 2. Components of IndVisSGG. (1) **Input Information.** As the blue part shown, the input information includes *Vision Information* (V_t) from videos along with predefined *Objects* (O), *Predicates* (P) and given *Examples* (E). E contains multiple positive and negative cases along with their respective analyses, concluding with an overall analysis. (2) **Prompt.** The *Prompt* consists of three parts: *INFORMATION*, *TRIPLETS EXTRACTION CRITERIA* (TEC) and *FORMAT*. *INFORMATION* specifies the meaning and characteristics of the scene graph. TEC outlines criteria for extracting entities. (3) **Multi-agent systems extract triplets.** The process is divided into three phases, with a total of 5 agents.

and relationships can lead to unsatisfactory outcomes. For instance, in some PSG datasets, the top three most frequent relationship categories account for over 50% of the samples, while many rare relationships occupy less than 1%, making them difficult to identify using PSG methods.

2.3. Large language model in manufacturing

Recently, LLMs with remarkable proficiency in few-shot reasoning and common-sense interpretation have gradually been deployed in various fields [26–28]. LLMs, along with their extensions such as VLMs, introduce novel opportunities for manufacturing tasks in the industry. For example, Zheng et al. [29] leveraged LLMs to generate robot task planning strategies from human instructions. Liu et al. [30] used LLMs to enable a robot to execute manipulations following human natural language prompts. Similarly, Wang et al. [31] introduced LLMs to navigate a mobile robot in industrial settings based on continuous human instructions.

Nevertheless, there is little work devoted to autonomous scenario understanding, such as SGG, which enables industrial spatial intelligence for flexible automation. Additionally, developing SGG datasets in industrial scenarios remains a time-consuming and resource-intensive process, primarily due to the challenge of reducing or eliminating the object annotation step, despite existing algorithms being able to recognize many objects and human actions. To the best of our knowledge, there has been no documented application of VLMs to the task of temporal SGG in industrial settings, suggesting unexplored potential in this area. Therefore, we investigate how VLMs could achieve a cognitive SGG task, resulting in the proposed IndVisSGG approach for smart manufacturing systems and other industrial tasks.

3. IndVisSGG: A VLM-based SGG method

The SGG task for videos in industrial settings involves a three-stage collaborative process among multiple AI agents developed based on GPT-4V, aligning with Chain-of-Thought (CoT) [32,33]. These agents can automatically optimize the quality of scene graphs and reduce hallucination problems through error correction, fine-adaptation, and summarization operations. As shown in Fig. 2, the processes include: (a) extracting spatial-temporal triplets from vision information, (b) triple-checking for oversights across temporal domains, and (c) summarizing spatial and interactive relations of triplets.

3.1. Extracting spatial-temporal triplets from vision information

Videos recorded in factories contain spatial-temporal relationships of industrial parts along production processes, represented as $G = (V, E, T)$, where V denotes subjects and objects, and E represents predicates over the time span T . To describe the video information, triplets structured as $\langle \text{subject}_t, \text{predicate}_t, \text{object}_t \rangle$ are introduced to extract semantic knowledge from a specific video frame. In this structure, the subject and object are entities representing the industrial parts, and the predicate describes their spatial and interactive relationships at the timestamp t . On this basis, the VLM function is designed and denoted as $VLM(\cdot)$, with the formula (1) representing the input to the VLM and the output from GPT-4V.

$$\text{out}_{\text{trpl}} = VLM(V_t, \text{Prompt}) \quad (1)$$

where V_t and *Prompt* represent vision information at the timestamp t and prompts for GPT-4V respectively.

LLMs like GPT-4V, even when given simple prompts such as “extract triplets from images and organize them in the $\langle \text{subject}, \text{predicate}, \text{object} \rangle$ format”, can generate numerous triplets. Due to the unrestricted nature of the extracted triplets, there is significant diversity in the

selection of entities and predicates. While these entities and predicates are similar to the ground truth, variations in their expressions can lead to suboptimal results.

The Triplets Extraction Criteria (TEC) includes pre-established object categories and predicate dictionaries, serving as reference knowledge for GPT-4V to generate triplets that accurately describe geometrical, interaction relationships between objects within the scene. First, TEC thoroughly annotates entities within sentences, avoiding pronouns and instead replacing them with specific, predefined object categories. TEC then enables VLMs to predict object relationships by selecting the most relevant terms from these predicate dictionaries, ensuring each triplet contains clear entities and predicates. The strategy of TEC can demonstrably improve the triplet extraction ability of VLMs. With objects O , predicates P , and examples E , the VLM function at a given timestamp t is designed as follows:

$$out_t^{s_1} = VLM(V_t, O, P, E, Prompt) \quad (2)$$

3.2. Triple-checking for oversights across temporal domains

The triples extracted from temporal video frames normally contain many biases and noise, as VLMs overlook predefined rules in many cases. For example, in Fig. 2a, the output of GPT-4V from step 1 may contain redundant entities like “wrench”, which is not shown in V_{t_1} . Additionally, nodes such as “terminals”, “beam”, and “panel” may be missing among the O , which influences triplet extraction in these frames. Lastly, although the predicate “taping” predicted in V_{t_3} similarly represents human actions as “knocking on”, it misaligns with the rules set in the TEC.

Therefore, in this stage, multiple AI experts are integrated to analyze the output from step 1, identifying and correcting omissions in the results based on TEC and examples E for temporal frames V_t . As depicted in Fig. 2b, $Expert_1$ accurately identifies the absence of “wrench_{t₁}” in the image (deleting redundant entities) and recovers missing or extra elements to ternary number in triplets of $\langle subject_i, predicate_i, object_i \rangle$. $Expert_2$ adds undetected triplets affected by O , such as “(terminals, on, conveyor)_{t₁}”, “(beam, on, robot arm)_{t₁}”, and “(panel, on, table)_{t₁}” to the results, adapting them to reflect the actual situation. $Expert_3$ correctly changes the imprecise description of “taping_{t₃}” to “knocking on_{t₃}” following TEC rules. In this way, multiple agents can identify different issues in the output from step 1 and provide analysis to improve the interpretability of triplet output for video input in different time frames. Each $Expert_i, i \in \{1, 2, 3\}$, can be represented in a formalized expression denoted as follows:

$$(out_t^{s_2}, a_i) = VLM(V_t, O, P, E, Prompt, out_t^{s_1}) \quad (3)$$

where a represents the expert’s analysis for Step 2.

3.3. Summarizing spatial and interactive relations of triplets

This step integrates and analyzes the outputs of the three experts from step 2 to determine the final spatial relations of objects in different video frames. The summary process is formalized as follows:

$$out_t^{s_3} = VLM(out_t^{s_2}, \alpha, Prompt), \quad (4)$$

where $\{out_t^{s_2}, \alpha\} = \{out_t^{s_2}, a_i \mid i \in \{1, 2, 3\}\}$

For the spatial relation interpretation of triplets, GPT-4V pays attention to the given O , P , and E , ensuring that all results are in a format that satisfies the rules specified by TEC. In this way, a temporal scene graph G is generated to illustrate the relationships of industrial parts in a video.

Examples of prompts in the bottom left corner of Fig. 2 illustrate those used in these three steps. Additionally, further prompts can be tailored based on these examples to apply IndVisSGG across different scenarios.

4. Industry scene graph dataset

Various datasets have been proposed over the past years for SGG tasks. These range from small-scale bounding box-based datasets like VRD [34] and UnRel [35], to the large-scale Visual Genome (VG) [22] dataset, and the panoptic segmentation-based PSG [24] dataset. Typically, these datasets focus primarily on everyday scenarios, lacking attention to industrial tasks such as human–robot interaction and manufacturing activities, where SGG plays an important role in robot decision-making based on linked spatial relationships between different objects. Besides, due to annotators’ limited expertise and diverse domain knowledge, some annotations fail to fully capture the necessary details for SGG tasks.

An ISG dataset is constructed for industrial scenarios and manufacturing activities involving multiple human and robotic agents. This construction follows three main steps.

Step 1. Data Collection. As shown in Fig. 3a, the data includes examples from a wide range of task scenarios, with the top 10 triplets labeled below. The ISG dataset is derived from several sources: the Cobots and Humans in Industrial Collaboration dataset (CHICO) [36], the Egocentric dataset from Ego4D [37], EPIC-KITCHENS [38], industrial parts assembly data from HA-ViD [39], furniture assembly data from IKEA [40], and robot visuomotor manipulation (RVM) data from CALVIN [41]. The data distribution is depicted in Fig. 3c.

Step 2. Concise Object Category and Predicate Dictionary. The VLM method is utilized to extract noisy scene graphs using only image inputs. Representative objects and predicates are then automatically extracted from these noisy scene graphs by the open-vocabulary LLMs.

Step 3. Rigorous Annotation Process. Manual annotation is employed to rigorously label the noisy scene graphs. The process includes: (1) filtering out incorrect triplets; (2) adding and modifying representative objects and predicates to align predefined objects and appropriate predicates, further refining the noisy scene graphs; and (3) manually verifying the refined scene graphs to produce the final ISG dataset. The frequency of entities in the ISG dataset is presented in Fig. 3b.

The ISG dataset boasts two key advantages. Firstly, the rigorous annotation process ensures accurate and comprehensive relationship annotations for all objects present in temporal scene images. As shown in Table 1, the ISG dataset comprises nearly 10,000 images from videos, featuring up to 688 objects and 144 predicates with appropriate granularity and minimal overlap. The word cloud of the predicate classes, shown in Fig. 4, indicates frequency through font size. Secondly, ISG is the first scene graph generation dataset to include industrial scenes, as depicted in Fig. 3a. The wide range of industrial task scenarios facilitates SGG’s understanding of tasks in production environments, such as human–robot collaboration and smart assembly.

5. Experiment

In this section, we first introduce the datasets, evaluation metrics, and parameter settings used in our experimental setup. Next, we present the comparison results of IndVisSGG with other SGG methods, as well as the demonstration of spatial–temporal scene graphs in industrial scenarios.

5.1. Experiment setup

5.1.1. Datasets

The VG dataset, PSG dataset, and ISG dataset are used in the experiment to evaluate SGG performance across various tasks. Firstly, **VG dataset** [22] is a widely used baseline dataset. The VG-150 variant [42, 43], which includes 150 object categories and 50 relation categories, is sufficient for SGG testing. To minimize overhead in the actual experiment process, a sampling strategy is adopted, where one image is selected out of every 26 from the valid testing set, resulting in approximately 1018 images for comparison with other methods, enabling quick

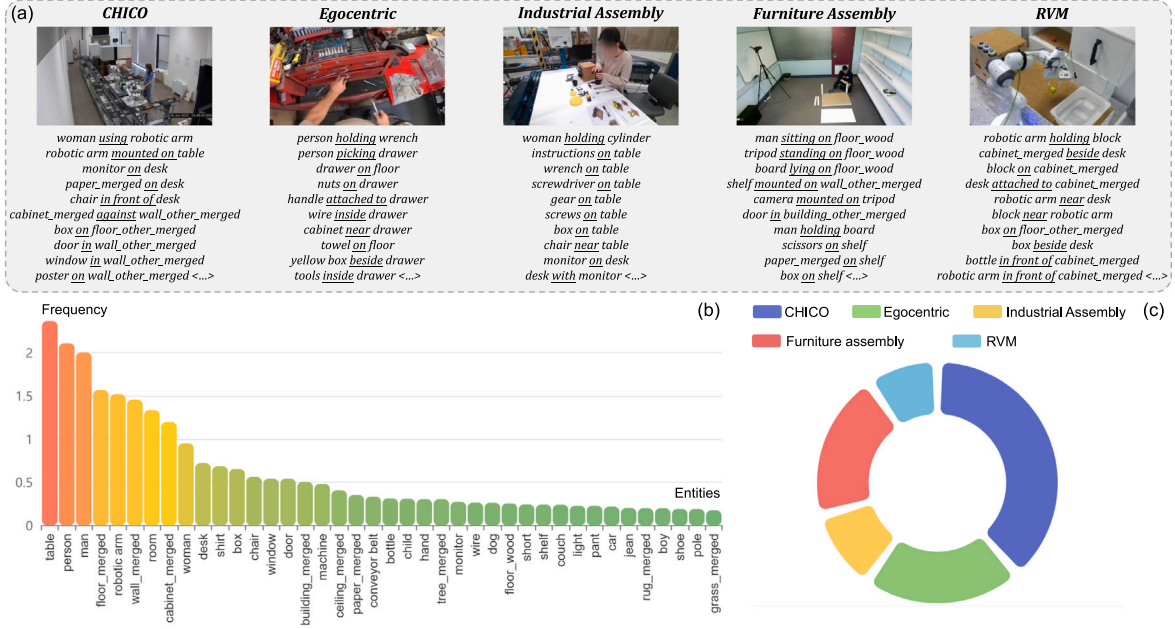


Fig. 3. ISG dataset scene distribution.

Table 1

Comparison between classic SGG datasets and ISG dataset. PPI counts predicates per image. DupFree checks whether duplicated object groundings are cleaned up. IIS indicates whether industrial scenes are included.

Dataset	Image	ObjCls	RelCls	PPI	DupFree	IIS	Source
VG	108K	34K	40K	21.4	✗	✗	COCO & Flickr
VG-150	88K	150	50	5.7	✗	✗	Clean VG
VrR-VG	59K	1600	117	3.4	✗	✗	Clean VG
GQA	85K	1703	310	50.6	✓	✗	Re-annotate VG
PSG	49K	133	56	5.7	✓	✗	Annotate COCO
ISG (ours)	10K	688	144	17.6	✓	✓	HA-ViD, IKEA, Ego4D, et al.



Fig. 4. Word cloud for ISG predicates.

manual assembly of mechanical parts, furniture assembly, robot autonomous manipulation, and human continuous actions in timestamped image frames of videos, as shown in Fig. 3a. Thus, the ISG dataset can be used to verify SGG performance on temporal images and demonstrate spatial-temporal scene graphs in sequential manufacturing tasks. The data distribution is shown in Fig. 3c, and entity frequency is depicted in Fig. 3b.

5.1.2. Metrics

Recall@K ($R@K$) and **meanRecall@K** ($mR@K$) are extensively utilized metrics in SGG tasks [24,42,44]. **Recall@K** measures the triplet recall and mean recall for each predicate category based on the top K triplets:

$$R@K = |Top_K \cap GT| / |GT|, \quad (5)$$

meanRecall@K calculates **Recall@K** for each predicate individually and then averages these values across all predicates.

5.1.3. Parameter settings

In the VG and PSG experiments, both O and P are predetermined for all methods. Unlike IndVisSGG, other methods require additional object annotations in the images using either bounding boxes or panoptic segmentation (like [45–47]). These traditional methods cannot be applied to the ISG dataset due to the lack of object annotations. To evaluate the ISG data, we introduced another advanced visual-language model, Gemini-Pro-Vision [48], incorporating *Prompt* and E as inputs. In all experiments, the number of IndVisSGG Experts was set to 3.

and effective assessment. Secondly, the **PSG dataset** [24] encompasses a large-scale collection of 48,749 annotated images, divided into 2186 images for testing and 46,563 images for training purposes. This dataset categorizes objects into 80 “thing” categories and 53 “stuff” categories, in addition to covering 56 categories of “relations”. We evaluate the PSG dataset using its testing set. To facilitate the evaluation process, the PSG dataset is split into training and testing sets in a ratio of 3:1. Lastly, the **ISG dataset** covers manufacturing-related scenes from a third-person perspective. It includes human-robot collaborative production,

Table 2

Results on VG, PSG, and ISG dataset. MOTIFS, FREQ, and VCTREE are reported from the re-implemented version [44].

Dataset	Method	Model input	Scene graph generation					
			R@20	mR@20	R@50	mR@50	R@100	mR@100
VG	FCSGG [45]	Vision	16.1	2.7	21.3	1.6	25.1	4.2
	MOTIFS [42]	Vision	21.4	6.6	27.2	8.2	30.3	15.3
	FREQ [42]	Vision	20.1	7.1	26.2	8.5	30.1	16.0
	RelTR [21]	Vision	21.2	6.8	27.5	10.8	—	—
	VCTree [44]	Vision	22.0	8.0	27.9	10.8	31.3	19.4
	IndVisSGG (ours)	Vision + Prompts	23.29[†]+1.29	12.98[†]+4.98	30.48[†]+2.58	21.39[†]+9.19	32.36[†]+1.06	21.99[†]+2.59
PSG	IMP [46]	Vision	17.9	7.35	19.5	7.88	20.1	8.02
	GPSNet [47]	Vision	18.4	6.52	20.0	6.97	20.6	7.17
	PSGFormer [24]	Vision	18.0	14.8	19.6	17.0	20.1	17.6
	MOTIFS [42]	Vision	20.9	9.60	22.5	10.1	23.1	10.3
	VCTree [44]	Vision	21.7	9.68	23.3	10.2	23.7	10.3
	IndVisSGG (ours)	Vision + Prompts	23.26[†]+1.56	16.25[†]+1.45	29.69[†]+6.39	25.12[†]+8.12	30.34[†]+6.64	25.62[†]+8.02
ISG	IndVisSGG-Gemini (ours)	Vision + Prompts	13.90	14.24	16.64	15.10	17.02	15.36
	IndVisSGG (ours)	Vision + Prompts	17.68	14.51	24.88	22.53	28.73	28.64

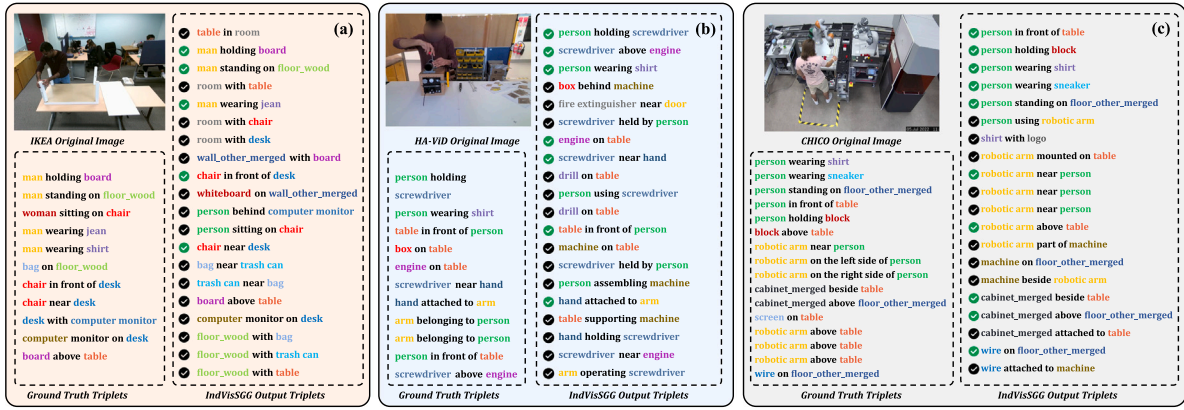


Fig. 5. Visualization of the top 20 triplets of three datasets predicted by IndVisSGG.

5.2. Comparison result and demonstration

5.2.1. Quantitative results

The comparative performance of IndVisSGG and other SGG methods across three datasets are presented in Table 2. For the ISG dataset, the “IndVisSGG-Gemini” refers to the integration of Gemini-Pro-Vision within IndVisSGG. Observations from these comparative results are as follows: (1) IndVisSGG achieved the highest R@K and mR@K scores on both the VG and PSG datasets compared to the other nine methods. This demonstrates the effectiveness of using VLMs to directly extract triplets from images.

(2) IndVisSGG significantly improves evaluation performance on the mR@K metrics, indicating its ability to effectively recognize rare categories in datasets, whereas traditional methods struggle with data distribution issues in such tasks. The TEC strategy enhances the accuracy of entity and predicate predictions across diverse scenarios.

(3) On the ISG dataset, IndVisSGG with GPT-4V agents outperforms Gemini-Pro-Vision agents in both R@K and mR@K. The performance scores, comparable to those of the other two datasets, suggest that the ISG data exhibits similar diversity in machinery information related to industrial tasks. This highlights its effectiveness in addressing the gap in industrial scene annotations within SGG datasets.

5.2.2. Qualitative results

The visual results of IndVisSGG on different datasets for SGG tasks are illustrated in Fig. 5. Reasonable triplets are indicated by ticks, with triplets matching the ground truth marked by green ticks. The visualization reveals that most triplets identified by IndVisSGG are present in the images and elaborate on the spatial and interactive relations of objects in these scenarios. For example, in the IKEA dataset (Fig. 5a),

IndVisSGG accurately predicted scenarios such as *man-holding-board*, *man-standing-on-floor wood*, and *chair-in front of-desk*, which align with the Ground Truth. Additional triplets, including *board-above-table* and *bag-near-trash* can be linked to describe real situations in the image, despite being missing from the Ground Truth. This indicates that the IndVisSGG method can capture more detailed information in industrial scenarios. Then, in Fig. 5b, depicting the assembly scenarios of the HA-ViD dataset, IndVisSGG can predict triplets such as *person-holding-screwdriver*, *screwdriver-above-engine*, and *engine-on-table*, accurately reflecting the Ground Truth and comprehending the spatial and interactive relations of these industrial parts in production. Additionally, IndVisSGG can generate triplets like *person-assembling-machine*, which provide high-level semantic information for a more comprehensive understanding of the task. Lastly, as shown in Fig. 5c for the human-robot collaboration in the CHICO dataset, IndVisSGG can reasonably identify various mechanical parts and their relationships in the shared workspace. The predicted triplets, including *person-in front of-table*, *person-holding-block*, and *robotic arm-mounted on-table*, indicate the relative spatial positions between robotic arms, the person, and objects in tasks. The triplet *person-using-robotic arm* identifies the interaction between human and robotic agents in manufacturing activities.

5.2.3. Ablation study

The ablation studies were conducted on 10% of the PSG testing dataset to illustrate the effectiveness of each component in IndVisSGG. Table 3 demonstrates the superior performance of IndVisSGG when fully equipped with all its components, particularly highlighting the significant role of *O*, *P*, examples, and analysis in improving precision. From the first line, it is evident that without any additional information,

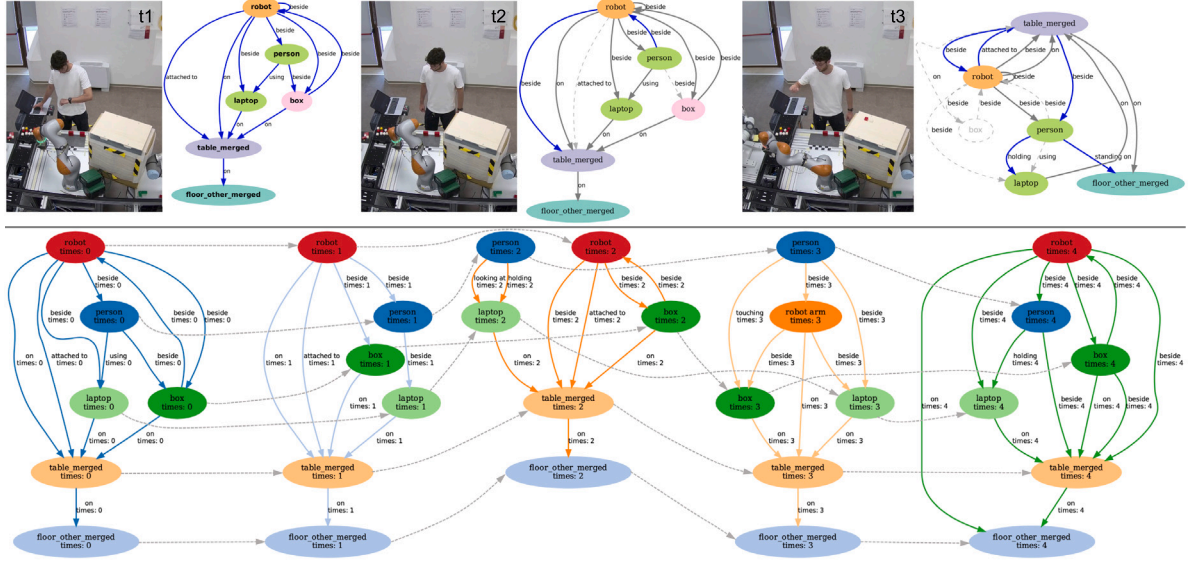


Fig. 6. Demonstration of scene graph along timesteps.

Table 3
Effects of different IndVisSGG components.

O	P	E&Analysis	R@20	mR@20	R@50	mR@50	R@100	mR@100
×	×	×	0.032	0.014	0.041	0.027	0.055	0.039
✓	×	×	1.787	0.378	3.195	2.645	3.272	2.652
×	✓	×	2.079	1.122	3.563	1.975	3.955	2.678
✓	✓	×	20.792	16.538	28.188	21.673	29.038	22.046
✓	✓	✓	23.040	17.250	29.785	27.375	30.147	27.680

Table 4
Ablation study on the number of experts.

Number	R@20	mR@20	R@50	mR@50	R@100	mR@100
1	21.323	15.400	23.530	18.701	24.431	24.920
2	21.584	16.428	25.371	19.247	26.541	25.396
3	23.158	16.947	28.447	25.480	30.142	27.031
5	23.287	17.890	28.754	24.383	30.020	27.591

IndVisSGG struggles to identify effective triplets. This is because, in the absence of *O* and *P*, the naming of objects and the expression of relations are unrestricted. The predicted triplets might be similar to the Ground Truth but differ in wording. Based on the results from the second and third lines, we observe that providing *P* alone leads to better outcomes than supplying *O* alone. This is primarily because predicting relations is often more challenging than forecasting object categories. From the fourth and fifth lines, it is shown that incorporating examples and analyzing those examples can effectively reduce the instances of IndVisSGG not adhering to TEC, significantly enhancing the accuracy of the results. As shown in Table 4, the performance of IndVisSGG improves as the number of experts increases. In our experiments, three experts proved to be the optimal number, achieving high accuracy in predicting triplets while minimizing token usage in VLM models.

5.2.4. Demonstration of spatial-temporal scene graph

The scene graph dynamically updates over time based on the video input, as presented in the upper part of Fig. 6. For temporally connected images, the dotted lines indicate disappeared edges and nodes, while the blue lines represent newly added edges and nodes. At timestamp t_1 , in human-robot collaboration scenarios, IndVisSGG identifies spatial and interactive relationships between different objects by linking triples such as *robot-beside-person*, *person-using-laptop*, *laptop-on-table*, and *robot-attached-to-table*. Then, at timestamp t_2 , the *beside* predicate between the *person* and *box* nodes is removed from the scene graph.

Finally, at timestamp t_3 , more spatial and interactive relations change, such as the disappearance of the *box* node and the addition of the *holding* predicate between the *person* and *laptop* nodes. Additionally, for continuous image frames in a video, a holistic scene graph is generated to capture the temporal information and changes in spatial and interactive relationships between objects, as shown in the bottom part of Fig. 6.

6. Conclusion

Scene graph generation is vital for industrial spatial intelligence, providing essential knowledge for seamless interaction and collaboration between human and robotic agents in smart manufacturing scenarios. To address the challenges of time-consuming and resource-intensive object annotation in previous SGG approaches, a novel VLM-based SGG method called IndVisSGG was proposed for spatial-temporal SGG in industrial settings. The IndVisSGG method consists of three steps: extracting spatial-temporal triplets from vision information, triple-checking for oversights across temporal domains, and summarizing spatial and interactive relations of triplets. All three steps were conducted using GPT-4V with effective prompts, eliminating the need for additional object annotations in images and demonstrating adaptability and generalization capability across diverse tasks. An ISG dataset was developed to support SGG tasks in industrial settings, including human-robot collaboration, robot manipulations, and assembly tasks in production. Through comparative experiments on the VG, PSG, and ISG datasets, the proposed IndVisSGG method was evaluated and found to outperform other SGG approaches in terms of accuracy. Additionally, spatial-temporal scene graphs were generated, demonstrating results in human-robot collaborative manufacturing tasks.

Despite the promising results, this work has some limitations. One limitation is that IndVisSGG was tested on VLM models such as GPT-4V and Gemini-Pro-Vision. The backbone of different VLMs may influence SGG performance on industrial tasks. In the future, more experiments will be conducted on robotics, allowing robots to utilize industrial spatial intelligence in SGG to coordinate, interact, and collaborate with humans in manufacturing tasks.

CRedit authorship contribution statement

Zuoxu Wang: Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Zhijie Yan:** Visualization, Methodology. **Shufei Li:** Writing – review & editing, Writing – original draft, Supervision, Methodology. **Jihong Liu:** Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This research work was funded by the grant from the National Natural Science Foundation of China (No. 52205244) and supported by the Ministry of Industry and Information Technology (MIIT), China Key Laboratory of Intelligent Manufacturing for High-end Aerospace Products, the Beijing Key Laboratory of Digital Design and Manufacturing, China, and the open program at the State Key Laboratory of Digital Manufacturing Equipment and Technology (No. IMETKF2024010), China.

Data availability

Data will be made available on request.

References

- [1] S. Li, P. Zheng, S. Liu, Z. Wang, X.V. Wang, L. Zheng, L. Wang, Proactive human-robot collaboration: Mutual-cognitive, predictable, and self-organising perspectives, *Robot. Comput.-Integr. Manuf.* 81 (2023) 102510.
- [2] S. Li, P. Zheng, J. Fan, L. Wang, Toward proactive human-robot collaborative assembly: A multimodal transfer-learning-enabled action prediction approach, *IEEE Trans. Ind. Electron.* 69 (8) (2022) 8579–8588, <http://dx.doi.org/10.1109/TIE.2021.3105977>.
- [3] L. Wang, A futuristic perspective on human-centric assembly, *J. Manuf. Syst.* 62 (2022) 199–201.
- [4] Z. Wang, X. Liang, M. Li, S. Li, J. Liu, L. Zheng, Towards cognitive intelligence-enabled product design: the evolution, state-of-the-art, and future of ai-enabled product design, *J. Clean. Prod.* 43 (2025) 100759.
- [5] J. Leng, Y. Zhong, Z. Lin, K. Xu, D. Mourtzis, X. Zhou, P. Zheng, Q. Liu, J.L. Zhao, W. Shen, Towards resilience in industry 5.0: A decentralized autonomous manufacturing paradigm, *J. Manuf. Syst.* 71 (2023) 95–114.
- [6] G. Agrim, S. Silvio, G. Surya, F.-F. Li, Embodied intelligence via learning and evolution, *Nat. Commun.* 12 (1) (2021) 5721.
- [7] S. Li, Y. You, P. Zheng, X.V. Wang, L. Wang, Mutual-cognition for proactive human-robot collaboration: A mixed reality-enabled visual reasoning-based method, *IIEE Trans.* 56 (10) (2024) 1099–1111.
- [8] L. Yang, H. Cai, Enhanced visual SLAM for construction robots by efficient integration of dynamic object segmentation and scene semantics, *Adv. Eng. Inform.* 59 (2024) 102313, <http://dx.doi.org/10.1016/j.aei.2023.102313>, URL <https://www.sciencedirect.com/science/article/pii/S147403462300441X>.
- [9] P. Zheng, S. Li, L. Xia, L. Wang, A. Nassehi, A visual reasoning-based approach for mutual-cognitive human-robot collaboration, *CIRP Ann.* 71 (1) (2022) 377–380.
- [10] T. Wang, J. Li, Z. Kong, X. Liu, H. Snoussi, H. Lv, Digital twin improved via visual question answering for vision-language interactive mode in human-machine collaboration, *J. Manuf. Syst.* 58 (2021) 261–269.
- [11] S. Li, P. Zheng, S. Pang, X.V. Wang, L. Wang, Self-organising multiple human-robot collaboration: A temporal subgraph reasoning-based method, *J. Manuf. Syst.* 68 (2023) 304–312.
- [12] J. Yang, C. Wang, B. Jiang, H. Song, Q. Meng, Visual perception enabled industry intelligence: State of the art, challenges and prospects, *IEEE Trans. Ind. Inform.* 17 (3) (2021) 2204–2219, <http://dx.doi.org/10.1109/TII.2020.2998818>.
- [13] L. Wang, R. Gao, J. Vánca, J. Krüger, X.V. Wang, S. Makris, G. Chrysosouris, Symbiotic human-robot collaborative assembly, *CIRP Ann.* 68 (2) (2019) 701–726.
- [14] W. Kim, L. Peternel, M. Lorenzini, J. Babič, A. Ajoudani, A human-robot collaboration framework for improving ergonomics during dexterous operation of power tools, *Robot. Comput.-Integr. Manuf.* 68 (2021) 102084.
- [15] S. Li, P. Zheng, L. Zheng, An AR-assisted deep learning-based approach for automatic inspection of aviation connectors, *IEEE Trans. Ind. Inform.* 17 (3) (2020) 1721–1731.
- [16] J. Xiao, J. Gao, N. Anwer, B. Eynard, Multi-agent reinforcement learning method for disassembly sequential task optimization based on human-robot collaborative disassembly in electric vehicle battery recycling, *J. Manuf. Sci. Eng.* 145 (12) (2023) 121001.
- [17] J. Fan, P. Zheng, S. Li, L. Wang, An integrated hand-object dense pose estimation approach with explicit occlusion awareness for human-robot collaborative disassembly, *IEEE Trans. Autom. Sci. Eng.* (2022).
- [18] S. Li, P. Zheng, J. Fan, L. Wang, Toward proactive human-robot collaborative assembly: A multimodal transfer-learning-enabled action prediction approach, *IEEE Trans. Ind. Electron.* 69 (8) (2021) 8579–8588.
- [19] J. Zhao, A. Giammarino, E. Lamon, J.M. Gandarias, E. De Momi, A. Ajoudani, A hybrid learning and optimization framework to achieve physically interactive tasks with mobile manipulators, *IEEE Robot. Autom. Lett.* 7 (3) (2022) 8036–8043.
- [20] Z. Yan, Z. Wang, S. Li, M. Li, X. Liang, J. Liu, ManufVisSGG: A vision-language-model approach for cognitive scene graph generation in manufacturing systems, in: 2024 IEEE 20th International Conference on Automation Science and Engineering, CASE, IEEE, 2024.
- [21] Y. Cong, M.Y. Yang, B. Rosenhahn, ReLTR: Relation transformer for scene graph generation, 2023, [arXiv:2201.11460](https://arxiv.org/abs/2201.11460).
- [22] R. Krishna, Y. Zhu, O. Groth, J. Johnson, K. Hata, J. Kravitz, S. Chen, Y. Kalantidis, L.-J. Li, D.A. Shamma, M.S. Bernstein, F.-F. Li, Visual genome: Connecting language and vision using crowdsourced dense image annotations, 2016, [arXiv:1602.07332](https://arxiv.org/abs/1602.07332).
- [23] Y. Liang, Y. Bai, W. Zhang, X. Qian, L. Zhu, T. Mei, VrR-VG: Refocusing visually-relevant relationships, 2019, [arXiv:1902.00313](https://arxiv.org/abs/1902.00313).
- [24] J. Yang, Y.Z. Ang, Z. Guo, K. Zhou, W. Zhang, Z. Liu, Panoptic scene graph generation, in: ECCV, 2022.
- [25] B. Cheng, I. Misra, A.G. Schwing, A. Kirillov, R. Girdhar, Masked-attention mask transformer for universal image segmentation, in: CVPR, 2022.
- [26] OpenAI, GPT-4 technical report, 2023, [arXiv:2303.08774](https://arxiv.org/abs/2303.08774).
- [27] B. Zhou, X. Li, T. Liu, K. Xu, W. Liu, J. Bao, CausalKGPT: Industrial structure causal knowledge-enhanced large language model for cause analysis of quality problems in aerospace product manufacturing, *Adv. Eng. Inform.* 59 (2024) 102333, <http://dx.doi.org/10.1016/j.aei.2023.102333>, URL <https://www.sciencedirect.com/science/article/pii/S1474034623004615>.
- [28] M.-C. Chiu, C.-Z. Tsai, Y.-C. Huang, Integrating object detection and natural language processing models to build a personalized attraction recommendation agent in a smart product service system, *Adv. Eng. Inform.* 61 (2024) 102484, <http://dx.doi.org/10.1016/j.aei.2024.102484>, URL <https://www.sciencedirect.com/science/article/pii/S1474034624001320>.
- [29] P. Zheng, C. Li, J. Fan, L. Wang, A vision-language-guided and deep reinforcement learning-enabled approach for unstructured human-robot collaborative manufacturing task fulfilment, *CIRP Ann.* (2024).
- [30] S. Liu, J. Zhang, L. Wang, R.X. Gao, Vision AI-based human-robot collaborative assembly driven by autonomous robots, *CIRP Ann.* (2024).
- [31] T. Wang, J. Fan, P. Zheng, An LLM-based vision and language cobot navigation approach for Human-centric Smart Manufacturing, *J. Manuf. Syst.* (2024).
- [32] J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. Chi, Q. Le, D. Zhou, Chain-of-thought prompting elicits reasoning in large language models, 2023, [arXiv:2201.11903](https://arxiv.org/abs/2201.11903).
- [33] M. Wu, J.-R. Lin, X.-H. Zhang, How human-robot collaboration impacts construction productivity: An agent-based multi-fidelity modeling approach, *Adv. Eng. Inform.* 52 (2022) 101589, <http://dx.doi.org/10.1016/j.aei.2022.101589>, URL <https://www.sciencedirect.com/science/article/pii/S1474034622000611>.
- [34] C. Lu, R. Krishna, M. Bernstein, F.-F. Li, Visual relationship detection with language priors, 2016, [arXiv:1608.00187](https://arxiv.org/abs/1608.00187).
- [35] A. Miech, J.-B. Alayrac, P. Bojanowski, I. Laptev, J. Sivic, Learning from video and text via large-scale discriminative clustering, 2017, [arXiv:1707.09074](https://arxiv.org/abs/1707.09074).
- [36] A. Sampieri, G. D'Amely, A. Avogaro, F. Cunico, G. Skenderi, F. Setti, M. Cristani, F. Galasso, Pose forecasting in industrial human-robot collaboration, 2022, [arXiv:2208.07308](https://arxiv.org/abs/2208.07308).
- [37] K. Grauman, A. Westbury, e.a. Eugene Byrne, Ego4D: Around the world in 3,000 hours of egocentric video, 2022, [arXiv:2110.07058](https://arxiv.org/abs/2110.07058).
- [38] D. Damen, H. Doughty, G.M. Farinella, S. Fidler, A. Furnari, E. Kazakos, D. Moltisanti, J. Munro, T. Perrett, W. Price, M. Wray, The EPIC-KITCHENS dataset: Collection, challenges and baselines, 2020, [arXiv:2005.00343](https://arxiv.org/abs/2005.00343).
- [39] H. Zheng, R. Lee, Y. Lu, HA-VID: A human assembly video dataset for comprehensive assembly knowledge understanding, 2023, [arXiv:2307.05721](https://arxiv.org/abs/2307.05721).
- [40] Y. Ben-Shabat, X. Yu, F.S. Saleh, D. Campbell, C. Rodríguez-Opazo, H. Li, S. Gould, The IKEA ASM dataset: Understanding people assembling furniture through actions, objects and pose, 2023, [arXiv:2007.00394](https://arxiv.org/abs/2007.00394).
- [41] O. Mees, J. Borja-Díaz, W. Burgard, Grounding language with visual affordances over unstructured data, 2023, [arXiv:2210.01911](https://arxiv.org/abs/2210.01911).
- [42] R. Zellers, M. Yatskar, S. Thomson, Y. Choi, Neural motifs: Scene graph parsing with global context, 2018, [arXiv:1711.06640](https://arxiv.org/abs/1711.06640).
- [43] L. Chen, H. Zhang, J. Xiao, X. He, S. Pu, S.-F. Chang, Counterfactual critic multi-agent training for scene graph generation, 2019, [arXiv:1812.02347](https://arxiv.org/abs/1812.02347).
- [44] K. Tang, H. Zhang, B. Wu, W. Luo, W. Liu, Learning to compose dynamic tree structures for visual contexts, in: Conference on Computer Vision and Pattern Recognition, 2019.
- [45] H. Liu, N. Yan, M.S. Mortazavi, B. Bhanu, Fully convolutional scene graph generation, 2021, [arXiv:2103.16083](https://arxiv.org/abs/2103.16083).
- [46] D. Xu, Y. Zhu, C.B. Choy, F.-F. Li, Scene graph generation by iterative message passing, 2017, [arXiv:1701.02426](https://arxiv.org/abs/1701.02426).
- [47] X. Lin, C. Ding, J. Zeng, D. Tao, GPS-Net: Graph property sensing network for scene graph generation, 2020, [arXiv:2003.12962](https://arxiv.org/abs/2003.12962).
- [48] Google, Gemini: A family of highly capable multimodal models, 2023, [arXiv:2312.11805](https://arxiv.org/abs/2312.11805).